

Documentation-Driven Threat Analysis

Research Work Plan

After documentation-layer closure – Revision 3

Systems-modeling prior-art gate before Base Analysis closure

Baseline. eb77d45edd09e3cd2959974bf57b1d066f7cbcbc.

Status. Chapters 2–4 CLOSED / FINAL; W0 CLOSED. BA0 is paused at entry while a bounded prior-art reading phase (BA0-R) tests systems-modeling, MBSE and architecture-description foundations. BA1 remains not started.

Worksheet policy. Reuse the existing `DDTA_modello_scheda_lettura_v2.pdf`; the reusable evidence record remains the Markdown source note plus YAML excerpt ledger.

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1 Current checkpoint

The governed documentation metamodel remains CLOSED for the current thesis scope through SecurityRequirement. Requirement remains the governed normative obligation; the separate L1 NormativeObligation wrapper remains rejected. SecurityRequirement retains its single protected security property, failure-mode explicitness and cause neutrality.

Chapter 2	CLOSED / FINAL
Chapter 3	CLOSED / FINAL
Chapter 4	CLOSED / FINAL
Documentation layer	CLOSED
WO	CLOSED
BA0 semantics	PAUSED AT ENTRY
BA0-R prior-art research	NEXT
BA1 BAE ontology	NOT STARTED

2 Literature workflow reused without a new worksheet family

BA0-R reuses the existing DDTA literature workflow. No second worksheet schema is introduced.

```
literature/worksheets/<SOURCE-ID>_scheda_compilata.pdf
literature/notes/<SOURCE-ID>.md
literature/excerpts/<SOURCE-ID>.excerpts.yml
```

The blank printable aid remains `literature/worksheets/DDTA_modello_scheda_lettura_v2.pdf`. The worksheet is a reading aid, not evidentiary authority; thesis claims must resolve to a registered source and an exact citation-ready location.

2.1 Canonical reading questions

1. What research problem does the source address and what contribution does it make?
2. Which artifacts does it start from and how does it assume, create, extract or derive a system representation?
3. How does it connect sources, requirements, evidence and change, and what is automated or human-reviewed?
4. What does it demonstrate, what are its limits, and what does it leave open for DDTA and the overlay method?

2.2 BA0-R extraction lens

The existing note may expand Question 2 and the DDTA relationship section using this phase-local lens:

```
SYSTEM REPRESENTATION
structure / part / component / participant
behavior / function / action
information / item / resource
flow / exchange
interface / port / endpoint
connection / channel / link
storage / persistence / memory
state / mode / condition
capability
```

```

environment / context
boundary / scope / responsibility
constraint / property
contract / schema
dependency / allocation

```

IDENTITY AND SEMANTIC ROLE

```

first-class identity vs role
entity vs relationship vs property
type vs usage / instance
independent mutation
ownership / allocation / composition

```

VIEWS AND ABSTRACTION

```

canonical model vs diagram
view / viewpoint / model kind
projection and omitted information
levels of abstraction

```

AUTHORITY AND EVOLUTION

```

source of truth
authored vs generated content
review / acceptance
versioning / traceability / change

```

DDTA CHALLENGE

```

reusable vs language-specific semantics
reinvention risk
gaps left open for documentation-driven analysis

```

Do not modify the general source-note template merely to add these prompts unless repeated use proves a genuine general deficiency.

3 BA0-R – Systems-modeling prior-art research gate

3.1 Purpose

Before defining Base Analysis responsibility or BAE taxonomy, determine how mature non-security modeling approaches represent systems, behavior, information exchange, interfaces, communication, persistence, boundaries, states, views and analysis-enabling semantics. The goal is prior-art falsification, not selection of a modeling language.

3.2 Non-goals

BA0-R does not define the final BAE metamodel; copy SysML/KerML/AADL/ArchiMate/NASA wholesale; turn Base Analysis into a DFD or threat model; promote current candidate terms to closed metaclasses; or treat bibliography mentions as verified sources.

Negative-control rule

The prior Actor + Asset + Interaction triad is no longer the default core. During BA0-R it is retained only as a negative-control hypothesis for sufficiency.

4 Fixed seed reading corpus

Provisional source IDs are planning reservations. Registry insertion occurs only after bibliographic identity and lawful access are verified.

Step	Source	Authority	Primary BA0-R question
BA0-R0	Existing SRC-0034, Souza, Moreira, Goulao (2019)	Existing secondary review	Re-anchor views, ADLs, human architectural decisions, representation-specific transformations and traceability limits. No duplicate source record.
BA0-R1 / SRC-0039	NASA-HDBK-1009A, <i>NASA Systems Modeling Handbook for Systems Engineering</i> (2025)	NASA official guidance	Which model elements/relations support SE work products, and how are diagrams/tables generated from the system model?
BA0-R2 / SRC-0040	OMG, <i>System Modeling Language (SysML) Specification</i> , v2.0 (2025)	Normative standard	Which structure, behavior, interface, flow, requirement, analysis and view concepts are first-class, and what is SysML-specific?
BA0-R3 / SRC-0041	OMG, <i>Kernel Modeling Language (KerML) Specification</i> , v1.0 (2025)	Normative kernel	What is the general semantic kernel beneath SysML, especially identity, feature, relationship, occurrence, behavior and connection semantics?
BA0-R4 / SRC-0042	NASA, <i>Systems Engineering Handbook</i> , NASA/SP-2016-6105 Rev2	NASA guidance	How are architecture, functions, flows, interfaces, resource-flow items, communication resources and technical-effort boundaries modeled outside security?
BA0-R5 / SRC-0043	ISO/IEC/IEEE 42010:2022, <i>Software, systems and enterprise – Architecture description</i>	International standard	How are architecture vs architecture description, viewpoints and model kinds separated? Full note only with lawful full-text access.
BA0-R6 / SRC-0044	Feiler, Gluch, Hudak (2006), <i>The Architecture Analysis & Design Language (AADL): An Introduction</i>	Primary technical report	Which component/interaction/software/hardware/bus/memory concepts enable repeated architectural analysis?
BA0-R7 / SRC-0045	The Open Group, <i>ArchiMate 3.2 Specification</i> (2022)	Standard	Does Active Structure / Behavior / Passive Structure provide a useful counter-model for DDTA hypotheses? Full note only under lawful license.
BA0-R8 / SRC-0046	Estefan (2008), <i>Survey of Candidate Model-Based Systems Engineering (MBSE) Methodologies</i> , Rev. B	INCOSE methodology survey	Which MBSE families and foundational references should enter the second reading ring, and where do representations diverge?

4.1 Source-access discipline

For every source: verify identity/version; record legal access; record local filename/hash when applicable; keep copyrighted PDFs out of Git unless redistribution is permitted; and never manufacture full-text claims from metadata. ISO 42010 and ArchiMate are explicit access gates: if lawful full text is unavailable, document the limitation and identify an open primary/authoritative replacement for substantive comparison.

5 Per-source deliverables and bibliography mining

For each new source that passes access verification:

<code>literature.registry.yml</code>	updated
<code>source-access.registry.yml</code>	updated
<code>literature/notes/SRC-XXXX.md</code>	completed
<code>literature/excerpts/SRC-XXXX.excerpts.yml</code>	completed
<code>literature/worksheets/SRC-XXXX_scheda_compilata.pdf</code>	generated
Follow-up sources	classified

Follow-up references are classified FOUNDATIONAL, COMPETING / ALTERNATIVE, or EMPIRICAL / EVALUATION. A follow-up receives a new source ID only after verification. Seed expansion is allowed only with an explicit rationale.

6 BA0-R cross-source synthesis and exit gate

Cross-source conclusions are written only after source-level reading, under `literature/syntheses/`. Required outputs cover common semantic responsibilities, divergent choices, entity/relationship/property/role distinctions, views/projections, boundaries, information paths, behavior, authority/evolution, bibliography expansion and a DDTA novelty challenge.

BA0-R exit gate

Every mandatory accessible seed source has a completed note, excerpt ledger and worksheet; access exceptions are explicit; a cross-source synthesis exists; no BAE metaclass was accepted merely because one notation contains it; and the novelty/reuse question was challenged. Any material Chapter 2/3 reopen trigger is reported before BA0 resumes.

7 Resume point after BA0-R

Unless BA0-R reveals a material falsification, execution returns to the Revision 2 sequence:

```
BA0 - Base Analysis responsibility and boundary
BA1 - Minimal BAE ontology
BA2 - Relations and canonical action vocabulary
BA3 - Document-to-BAE derivation, provenance and authority
BA4 - Multi-level views and projections
BA5 - Controlled vocabulary and authoring assistance
BA6 - Base Analysis closure and regression
```

8 BA0 – Base Analysis responsibility and boundary

After BA0-R, define Base Analysis as a governed, methodology-neutral analyzable representation grounded in governed documentation and explicit reviewed additions. It is not a second narrative hierarchy, a STRIDE DFD, or automatically canonized NLP/LLM extraction. One canonical knowledge base may support several projections, methods cannot redefine its semantic core, and the representation must justify why its responsibilities are not better delegated to an adopted standard/modeling language.

BA0 exit gate

A precise responsibility statement and non-goals are accepted after prior-art synthesis and before BAE taxonomy design.

9 BA1 – Minimal BAE ontology

Start from semantic responsibilities forced jointly by the governed DDTA corpus and BA0-R evidence, not from a pre-approved type list. Actor + Asset + Interaction is a sufficiency negative control. Structural participant, behavior/function/action, information/resource item, flow/exchange, interface/endpoint, connection/channel, persistence/storage, state/mode, capability, environment/context and boundary remain hypotheses until falsified.

BA1 exit gate

Minimal generic BAE set with definitions, identity/ownership rules, relation/property boundaries and negative controls.

10 BA2–BA6 summary

BA2: close the smallest stable relation vocabulary and separate semantic relation, canonical lexical label and authoring suggestion.

BA3: distinguish governed fact, candidate extraction, reviewed canonical BAE and reviewed analytical addition; close provenance and authority.

BA4: define multiple views as projections of one canonical model, with at least two consistent facial-access views under mutation.

BA5: define controlled-vocabulary/LLM assistance without semantic authority.

BA6: regress corpora/holdout, identity stability, mutations, view consistency, method neutrality and insufficient-information diagnostics.

Base Analysis exit gate

A methodology-neutral Base Analysis can be built and reviewed from governed documentation plus explicit reviewed additions, with stable identities, relations, provenance and projections, and with its relationship to existing systems-modeling prior art made explicit.

11 Generic analysis envelope and overlays

Only after the Base Analysis gate: A1 AnalysisRecord, A2 Finding, A3 change/provenance integration, O1 generic overlay contract, O2 STRIDE, O3 closed-loop improvement and O4 STRIDE-AI. Later use LINDDUN/another method as neutrality challenges. ThreatForge remains an implementation/tooling case study and never semantic authority.

12 Next authorized microstep

Authorized next step

Execute only **BA0-R1**: read NASA-HDBK-1009A deeply enough to populate the existing DDTA source-note, excerpt and worksheet workflow, then mine its bibliography. Before that reading, reread the existing SRC-0034 note as the bridge source. Do not start BA1, STRIDE, AnalysisRecord, Finding or implementation work while BA0-R is open.